

Unequal Consumption and Savings in the Heterogeneous-Agent Model

Pascal Michailat
<https://pascalmichailat.org/w/>

Household i:

Purchases of services:

$$y_i = \sigma(x) \left[f(x) k_i + \frac{p_i}{p} \right]$$

Marginal propensity
to spend (MPS)

where $\sigma(x) = \frac{x^\varepsilon [1 + \tau(x)]^{1-\varepsilon}}{1 + x^\varepsilon [1 + \tau(x)]^{1-\varepsilon}}, \sigma(x) \in (0, 1)$

Real wealth holdings (savings)

$$\frac{m_i}{p} = [1 - \sigma(x)] \left[f(x) k_i + \frac{p_i}{p} \right]$$

Marginal
propensity to
save

Consumption:

$$c_i = y_i / [1 + \tau(x)]$$

Visits:

$$v_i = y_i / q(x)$$

Household i with more initial wealth (p_i)

- More consumption (c_i)
- More purchases/expenditure (y_i) $\swarrow$ of services
- More visits (v_i)
- More final real wealth/savings (m_i/p)

Same results for household i with higher income (k_i)